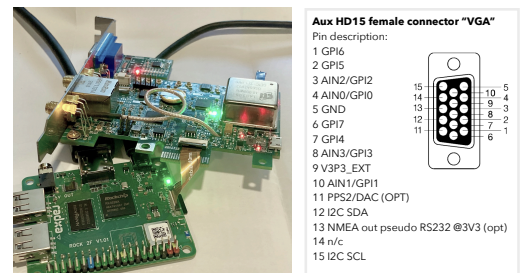
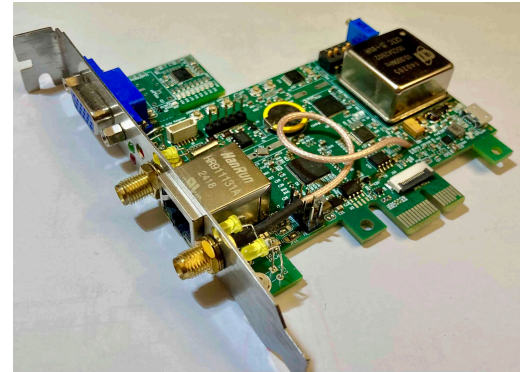
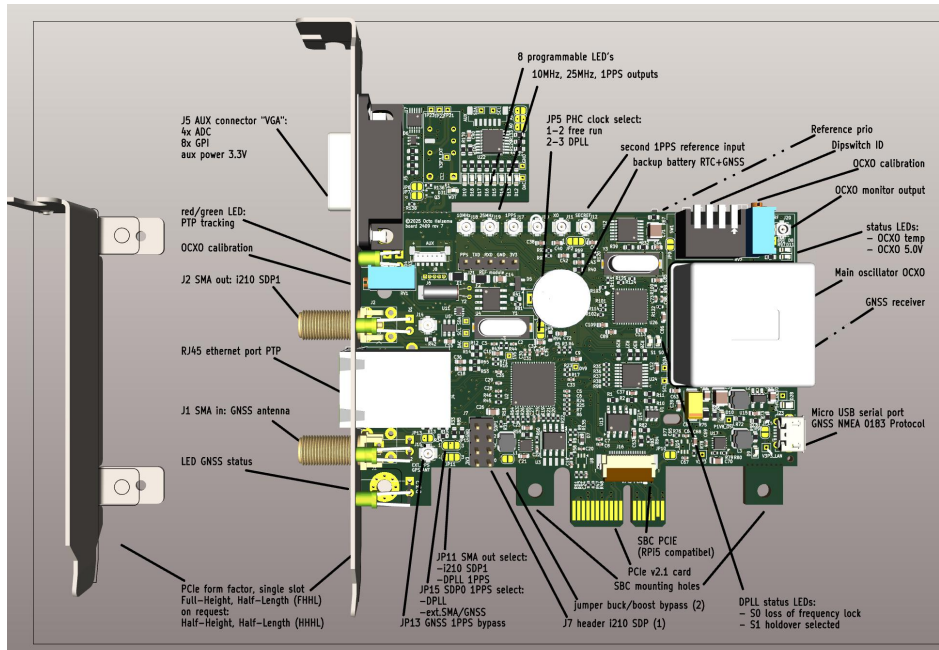


PCIe NIC with onboard GNSS for PC and Raspberry Pi 5

Model: Board 2409-7
Designed by: Octo Halsema

Product brief - PB2409-02 - May 2026



Aux HD15 female connector "VGA"	
Pin description:	
1	GPi6
2	GPi5
3	AIN2/GPi2
4	AIN0/GPi0
5	GND
6	GPi7
7	GPi4
8	AIN3/GPi3
9	V3P3_EXT
10	AIN1/GPi1
11	PPS2/DAC (OPT)
12	I2C SDA
13	NMEA out pseudo RS232 @3V3 (opt)
14	n/c
15	I2C SCL

Description

Network interface card with GNSS disciplined oscillator based on Intel i210 1Gbe with IEEE1588 PTP Hardware Clock (PHC). PTP core and PHY clocked from DPLL jitter cleaner with ovenized local oscillator for zero ppm frequency offset. SDP0 (1) pin internally connected to 1PPS. USB serial port to access NMEA protocol and 1PPS on carrier detect (CD) line. All trivial functions supported by upstream linux kernel and win-xp,7,8,10,11 drivers "igb" and "ch343", for x86_64 and ARM operating systems. Patched linux driver needed for extra functions: DPLL kernel messages, RTC, ID dipswitch, GPIO, ADC and status leds. This board can also run without computer (PC or RPi) for pure GPSDO functionality, powered from 3.3-5V 8pin header. Comes with default frequencies pre-programmed 1PPS+10MHz+25MHz.

Applications

Packet analyzer, data acquisition and measurement, GPI event timestamping, frequency measurement, NTP+PTP server, time-sensitive networking, audio-video bridging, GPS logger.

I look forward to see your unique use-cases for this unconventional device: a networking card combined with a GPSDO. Octo Halsema

Key features

PTP Hardware Clock

Mode:	IEEE 1588v2 leader/follower
Local oscillator:	OCXO temp.stab. 5x10-8
DPLL:	1,5 PPM lock range, LBW 0.01 Hz
Ethernet:	PHC/PHY clocked from DPLL (SyncE master)

Reference

GNSS:	GPS/BDS constellation, SMA antenna bias 3.3V
1PPS input:	SMA connector external
PTP:	PTP Hardware Clock

Signal inputs/outputs

Clock out:	U.fl connector, LVCMOS level, 50 Ohm
Frequency:	1PPS, 10MHz, 25MHz simultaneously
SDP1 (1):	SMA connector in/out, precise arbitrary frequency
Micro USB:	GNSS NMEA serial port (CH343 chip)
Auxiliary HD15:	8x GPIO PCF8574, 4x ADC ADS1115, +3.3V
Ethernet RJ45:	IEEE 802.3 Ethernet interface for 1000BASE-T, 100BASE-TX, and 10BASE-T

Power supply

Converter stabilizer (2):	buck/boost converter for main 3.3V LDO stabilizers for PLLs
Sources:	PCIe bus 3.3V RPi5 16-way FCC connector 5.0V pin header 3.3-5.0V

(1) SDPn = Intel i210 Software Definable Pin

(2) Power consumption depends on ports used/enabled, possibly exceeding converter's capability especially at low voltage (3.3V). Caution when using buck/boost bypass!